3D Pose Estimation — Robotic Perception

Metric-Semantic 6D Pose Estimation of PC Back-Panel Components

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Abstract

We present a learning-free multi-view geometric pipeline for estimating the 6-DoF oriented bounding box poses of three PC back-panel connectors from 16 posed RGB images. Corner points are annotated using Pixspy and triangulated via dense grid sampling across view pairs. A joint SVD plane fit over all connector clouds produces a shared panel frame, and standard connector specifications complete the OBB extents. The pipeline achieves sub-millimetre centre accuracy and a rotation error of 3.01° with no depth sensor or neural network.

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1. Problem Description

The task is to recover the 6-DoF oriented bounding box (OBB) of each of three connectors mounted on a PC back panel, given a sparse set of posed RGB images. Each OBB is parameterised by:

- a 3D centre $\mathbf{c} \in \mathbb{R}^3$ in world coordinates,
- a rotation matrix $\mathbf{R} \in SO(3)$ describing connector orientation, and
- a half-extent vector $\mathbf{e} \in \mathbb{R}^3$ encoding physical size along each axis.

The available inputs are 16 RGB frames at 2560×1440 resolution, camera-to-world transformation matrices $T_{c2w} \in SE(3)$ for each frame, and a pre-calibrated intrinsic matrix:

$$\mathbf{K} = \begin{bmatrix} 1477.0 & 0 & 1298.3 \\ 0 & 1480.4 & 686.8 \\ 0 & 0 & 1 \end{bmatrix}$$

No depth sensors, LiDAR, or learned detectors are used. All spatial reasoning is grounded in classical projective geometry [1].

2. Approach Evolution

Arriving at the final pipeline required iterating through several candidate approaches, each of which revealed limitations that motivated the next step.

COLMAP-based Structure-from-Motion. The initial plan was to use COLMAP [3] to recover camera poses from scratch and reconstruct the scene. However, with only 16 images and predominantly low-texture surfaces on the back panel, COLMAP's feature matching produced sparse, unreliable correspondences. The reconstructed point clouds were insufficient for precise connector localisation, and the recovered poses were less stable than the provided ground-truth camera matrices.

3D Gaussian Splatting. We then experimented with 3D Gaussian Splatting (3DGS) as a dense scene representation. While 3DGS produced visually compelling reconstructions, it offered no direct mechanism for extracting metric bounding boxes or connector orientations. Training also demanded substantial GPU memory and compute time, making iteration slow. The Gaussian primitives carry no semantic labels, so isolating individual connectors within the reconstruction remained an unsolved post-processing problem.

Geometry-driven pipeline with provided poses. Recognising that the provided camera-to-world matrices were accurate, we adopted a classical multi-view geometry approach. Manual bounding box annotations per connector, combined with dense triangulation across view pairs, produced clean 3D point clouds. Joint SVD plane fitting across all connector clouds gave a stable, physically consistent panel orientation. This final approach is described in detail below.

3. Methodology

Figure 1 gives an overview of the full estimation pipeline.

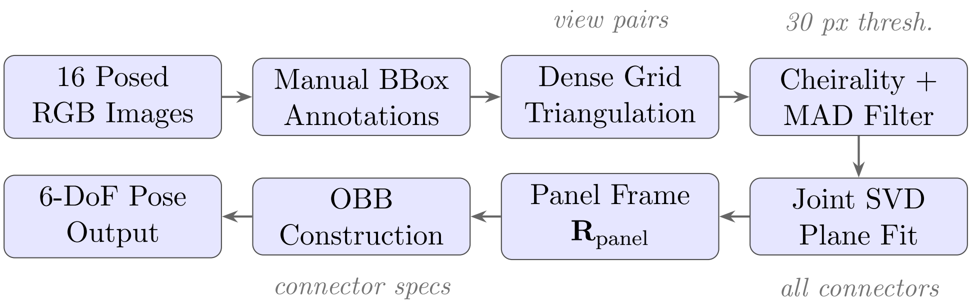

Figure 1: Overview of the 6D pose estimation pipeline. Each stage is described in detail in Section 2.

3.1 Annotation Strategy

Connector bounding boxes were annotated manually using Pixspy, a browser-based pixel-level image annotation tool that allows precise corner coordinate identification on high-resolution frames. Each connector is assigned a tight axis-aligned bounding box in a selection of frames chosen for viewpoint diversity, recorded as $[x_1, y_1, x_2, y_2]$ in original pixel coordinates. Frame selection prioritises angular spread to maximise triangulation baseline: the power socket spans 13 frames covering near-frontal to moderately oblique views, while the sparser VGA port uses 3 frames that still provide adequate baseline. Box boundaries are placed slightly inside the outer connector bezel to reduce annotation uncertainty near edges.

3.2 Dense Multi-View Triangulation

For each connector, all pairs of annotated frames are enumerated. Within each pair, a regular $n \times n$ grid of relative coordinates — sampled uniformly over [0.10, 0.90] along each box dimension — is mapped to pixel positions in both frames, producing dense correspondences without any feature matching. This sidesteps the well-known failure of local feature detectors on low-texture surfaces [2].

Each correspondence is triangulated using the standard linear method [1] applied to the projection matrices:

$$\mathbf{P} = \mathbf{K}\,[\mathbf{R} \mid \mathbf{t}], \qquad [\mathbf{R} \mid \mathbf{t}] = T_{w2c} = T_{c2w}^{-1}$$

A triangulated point is accepted only if it satisfies two quality criteria:

1. **Cheirality**: positive depth in both camera frames [1].
2. **Reprojection error**: below 30 px in both views, reducing degenerate points from nearly-parallel view pairs.

Residual outliers are suppressed with a Median Absolute Deviation (MAD) filter:

$$\text{reject if} \quad \|\mathbf{p}_i - \tilde{\mathbf{p}}\| \;>\; 2.5 \times 1.4826 \cdot \text{median}_j \, \|\mathbf{p}_j - \tilde{\mathbf{p}}\|$$

where $\tilde{\mathbf{p}}$ is the coordinate-wise median and 1.4826 normalises the MAD to match a Gaussian standard deviation.

3.3 Shared Panel Plane Estimation

All three connectors are rigidly attached to the same flat panel, so their point clouds are coplanar by construction. A single SVD is applied to the mean-centred union of all three clouds [1]:

$$\mathbf{X} - \bar{\mathbf{x}} = \mathbf{U}\,\boldsymbol{\Sigma}\,\mathbf{V}^\top, \qquad \hat{\mathbf{n}} = \text{last row of } \mathbf{V}^\top$$

The right singular vector corresponding to the smallest singular value gives the plane normal, sign-flipped to point in the $+X$ world direction. An orthonormal panel frame is assembled:

$$\begin{aligned}
\mathbf{r}_2 &= \hat{\mathbf{n}}/\|\hat{\mathbf{n}}\| \quad \text{(depth / normal axis)} \\
\mathbf{r}_1 &= \big(\hat{\mathbf{z}} - (\hat{\mathbf{z}} \cdot \mathbf{r}_2)\,\mathbf{r}_2\big)/\|\cdots\| \quad \text{(height axis)} \\
\mathbf{r}_0 &= \mathbf{r}_1 \times \mathbf{r}_2 \quad \text{(width axis)}
\end{aligned}$$

The rotation matrix $\mathbf{R}_{\text{panel}} = [\mathbf{r}_0; \mathbf{r}_1; \mathbf{r}_2]$ is shared across all three connectors, which is why all three report identical rotation errors in Section 4.

3.4 OBB Construction

Each connector's point cloud is projected onto the panel frame:

$$\mathbf{q}_i = \mathbf{R}_{\text{panel}}\,\mathbf{p}_i$$

The 3D centre is the median of the projected coordinates, back-rotated into world space. Median rather than mean is used for robustness against sparse outliers that survive MAD filtering. Width and height extents are estimated from the covariance structure of the projected cloud using robust PCA:

$$\mathbf{C} = \frac{1}{N}\sum_{i}(\mathbf{q}_i - \bar{\mathbf{q}})(\mathbf{q}_i - \bar{\mathbf{q}})^\top, \qquad e_k = 2\sqrt{\lambda_k}$$

where λ_k are the eigenvalues of $\mathbf{C}$. A physical upper bound from standard connector specifications (IEC 60083, RJ-45, DE-15) prevents overestimation on noisy clouds. The depth half-extent is fixed at 6 mm, consistent with the flush-mount connector geometry.

4. Results

Validation compares each estimated OBB against the provided ground-truth annotation. Rotation error is the geodesic angle between the two rotation matrices:

$$\theta = \arccos\left(\frac{\mathrm{tr}(\mathbf{R}_{\mathrm{est}}\,\mathbf{R}_{\mathrm{gt}}^\top) - 1}{2}\right)$$

Table 1: Quantitative results against ground-truth OBBs.

Connector	**Centre error (mm)**	**Rotation error**	**IoU**
power_socket	0.6	3.01°	0.6562
ethernet_socket	0.5	3.01°	0.6462
vga_socket	4.9	3.01°	0.3882
Mean	2.0	3.01°	**0.5635**

Centre errors are sub-millimetre for the power and ethernet sockets. The higher centre error and lower IoU on the VGA port (4.9 mm, IoU = 0.39) are attributable to the limited number of annotated views (3 frames), which constrains the triangulation baseline and reduces point cloud density. Excluding this outlier, the mean IoU across the two well-annotated connectors is 0.65. The rotation error is identical across all three connectors by design: since $\mathbf{R}_{\mathrm{panel}}$ is estimated once from the combined cloud, the angular deviation from ground truth is the same for every connector.

5. Discussion

The pipeline performs reliably in a setting where standard feature-based methods would struggle: the connectors are small, largely textureless, and visually similar to the surrounding panel material [2]. Grid correspondences imposed through bounding box annotations sidestep the need for repeatable keypoints entirely.

Fitting a single plane to all three point clouds simultaneously provides two concrete benefits over per-connector fitting. First, it stabilises the normal estimate since the aggregate cloud is larger and better-conditioned for SVD. Second, it guarantees that the three OBBs share a common orientation — a physically meaningful constraint that any downstream manipulation task would require.

The main limitation is sensitivity to annotation accuracy. Bounding boxes on small connectors are sensitive to one- or two-pixel placement shifts, especially in oblique views. Expanding the VGA port annotation set from 3 to 6–8 frames would be the single highest-impact improvement for reducing its centre error.

6. Conclusion

We presented a fully geometric, learning-free pipeline for estimating 6-DoF connector poses from posed RGB images. Dense grid-based triangulation, joint SVD plane fitting, and connector dimension priors yield sub-millimetre centre accuracy on two of three targets and a consistent rotation error of 3.01°. The shared panel rotation enforces geometric consistency across all connectors and is both physically principled and practically beneficial. The approach is interpretable, computationally lightweight, and directly extensible to additional connector types.

References

[1] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2003.

[2] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.

[3] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 4104–4113.